Workshop: AI Adoption in Data Analytics

Support by grant “AI Adoption in Data Analytics”

Xiaorui Zhu

Agenda

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Introduction to Generative AI

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Generative AI Applications

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What's New with GenAI

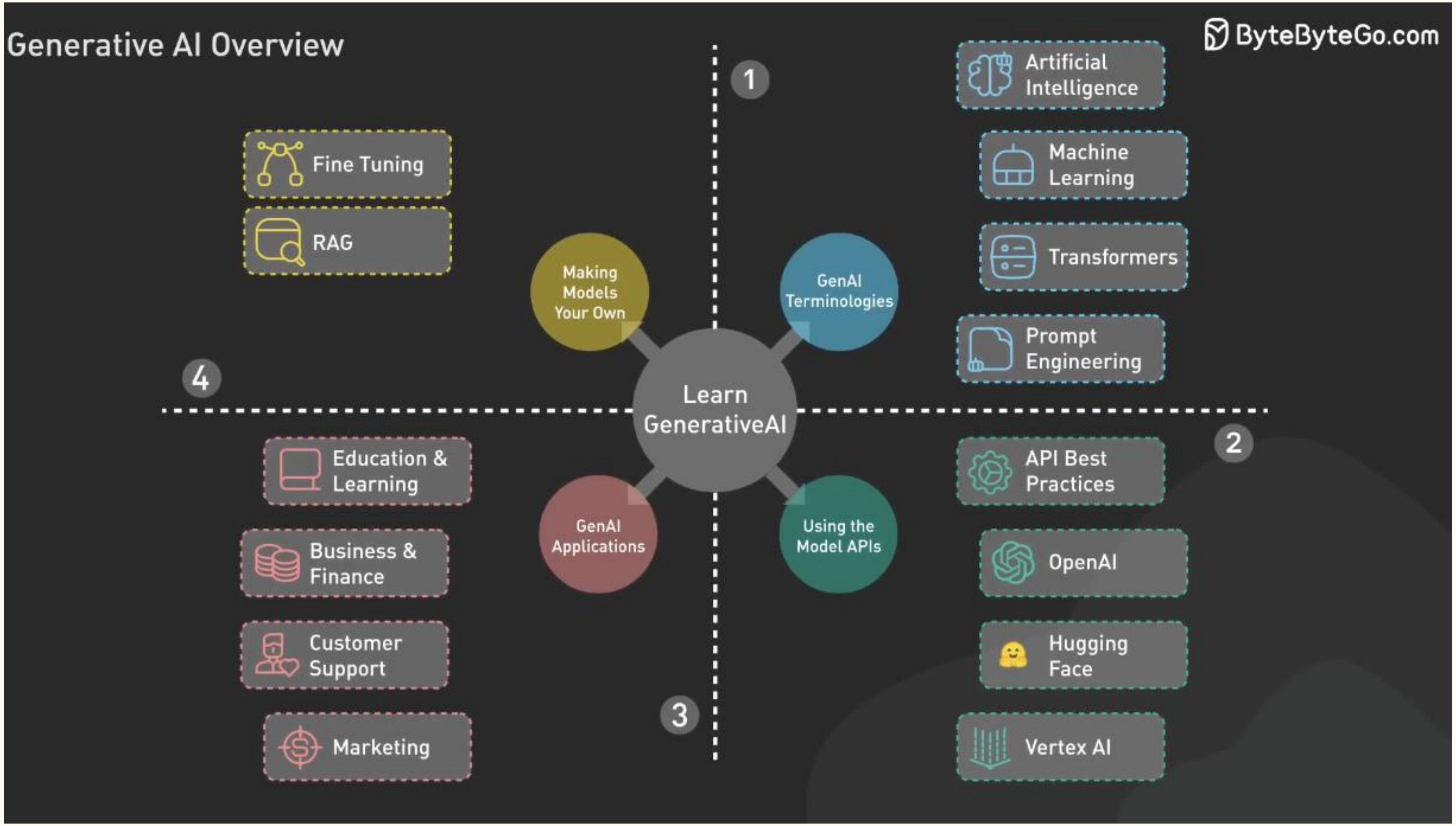
04

How Large Language Models Work

How Many of You Use ChatGPT?

- [ ] Ever?
- [ ] Every Week?
- [ ] Every Day (Almost)?

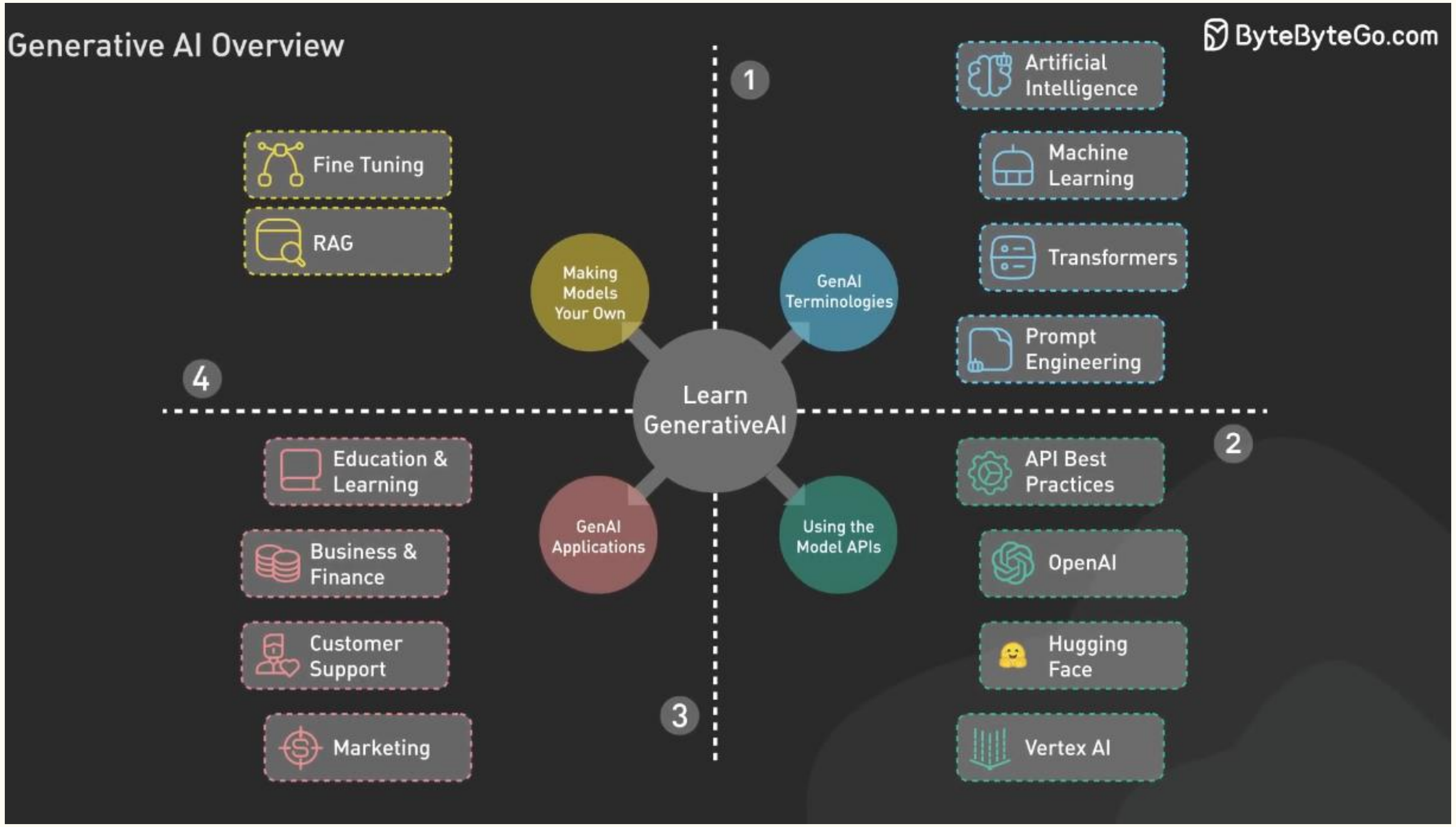
Introduction to Generative AI



https://bytebytego.com/

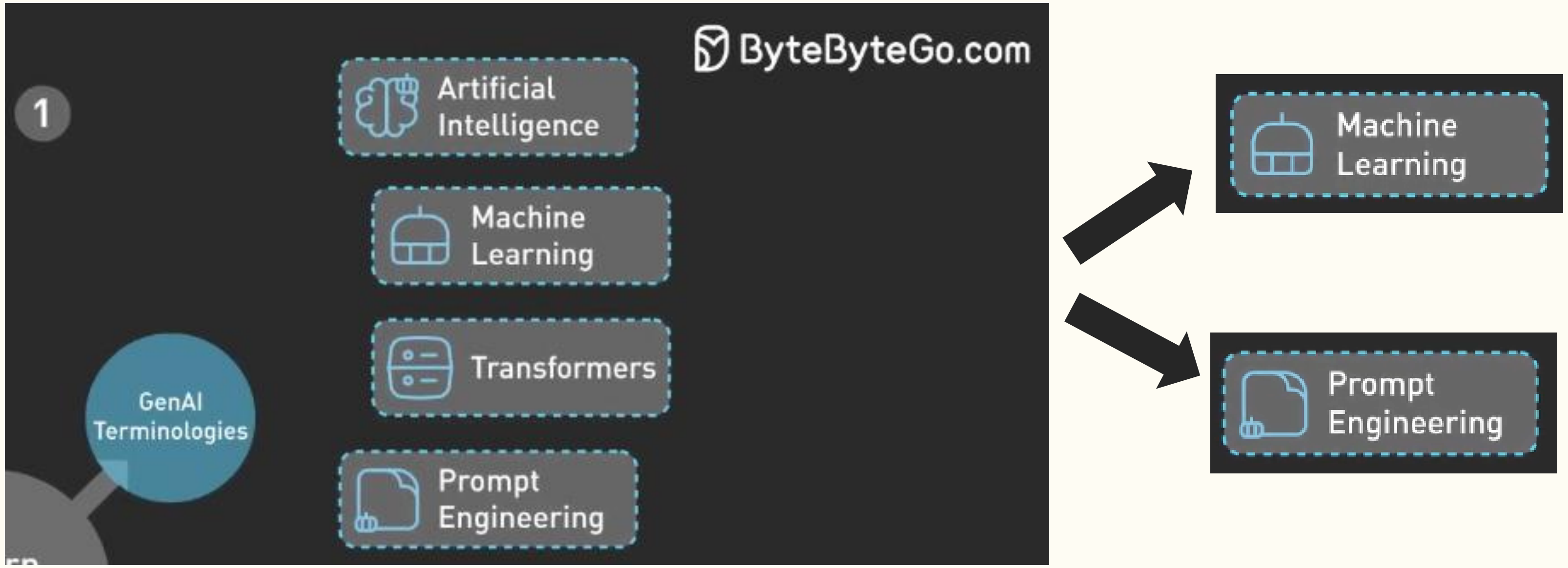
The Four Pillars of GenAI (The Big Picture)

- **Quadrant 1:** Terminologies & Core Concepts
- **Quadrant 2:** Leveraging Model APIs
- **Quadrant 3:** Industry-Specific Applications
- **Quadrant 4:** Customization & Fine-Tuning



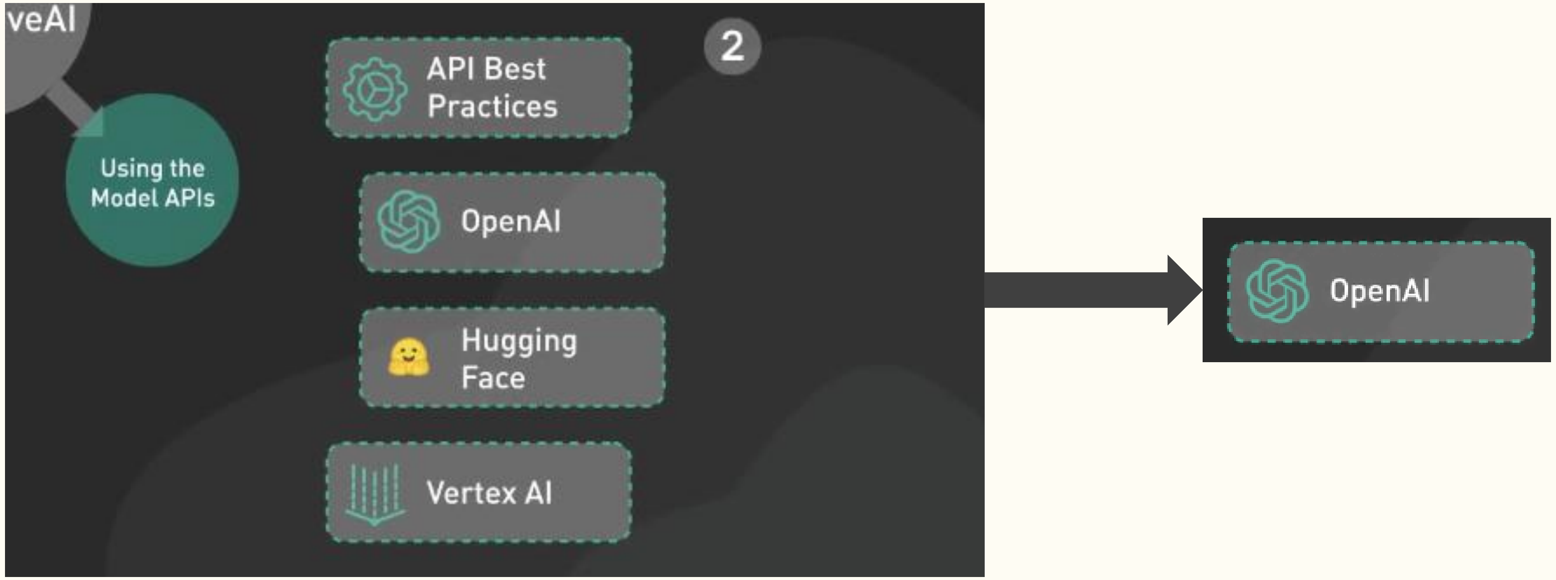
Pillar 1 – Foundations & Terminology

- **Key Concepts:**
  - **Artificial Intelligence:** The broad field of creating "smart" machines.
  - **Machine Learning:** Systems that learn from data and can evolve.
  - **Transformers:** The specific architecture powering modern LLMs.
  - **Prompt Engineering:** The art of crafting inputs to get better outputs.



Pillar 2 – Using Model APIs

– **OpenAI**: Prompt engineering on OpenAI
– **Hugging Face:** The "open-source" hub for pre-trained models.
– **Vertex AI:** Google’s enterprise-grade AI platform.
– **API Best Practices:** Focus on API security, latency, and cost management.



Pillar 3 – Generative AI Applications

– **Education**: Personalized learning paths and content creation.
– **Business/Finance:** Automating reports and trend analysis.
– **Customer Support:** 24/7 intelligent chatbots.
– **Marketing:** Copywriting, image generation, and personalization.
•

Generative AI

Artificial intelligence systems that can produce high-quality content using natural language

Text

Generate written content from articles and reports to code and creative writing

Image/Audio/Video

Create visual and audio content including photos, videos, illustrations, and music

Code

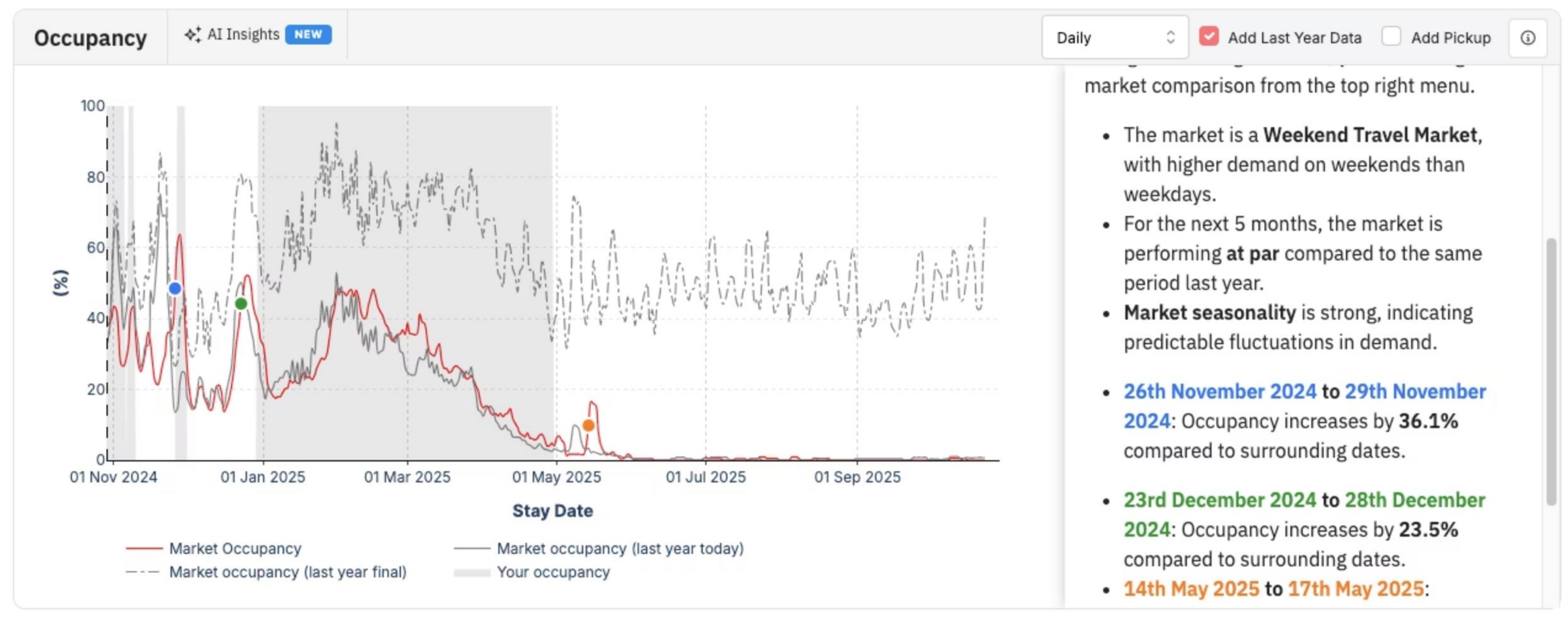
Write, debug, and explain software across multiple programming languages

GenAI Can See the World Now

David Attenborough is Now Narrating My Life

> https://www.youtube.com/watch?v=wOEz5xRLaRA

Into Products!

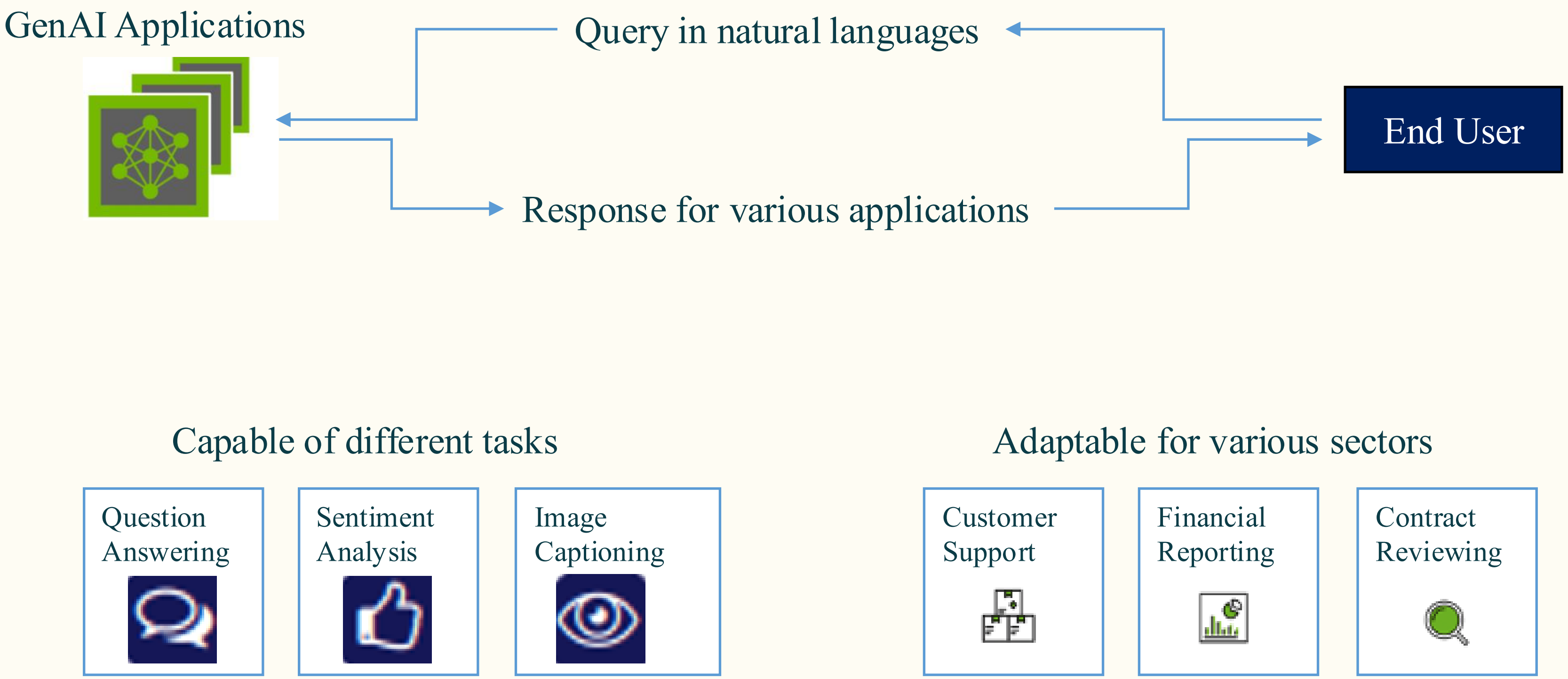


What began as experimental AI capabilities have rapidly transformed into production-ready products and services.

Major technology companies and startups alike are racing to integrate generative AI into their core offerings, fundamentally changing how users interact with software and services.

GenAI Applications

The Model of Interactions with GenAI



Pillar 4 – Making Models Your Own

- **The Two Methods:**
  - **Fine Tuning:** Adjusting model weights on specific datasets for niche expertise.
  - **RAG (Retrieval-Augmented Generation):** Connecting the model to your own live data for factual accuracy.

Use cases

- **Analytical product:** prototyping analytical tools
- **Local "AI chatbot":** Open WebUI is an extensible, self-hosted AI interface that operates entirely offline.
- **Research using local LLM**
- **Local LLM server:** for legal documents or sensitive information
- **Private AI Agents**: Link

What's New with GenAI? A Brief History

1950s: Symbolic AI

> "A physical symbol system has the necessary and sufficient means for general intelligent action"
>
> — *Newell and Simon, 1976*

Symbolic AI dominated the AI research landscape from the mid-1950s through the middle 1990s. This paradigm was built on high-level symbolic (human-readable) representations of problems, leveraging logic and search algorithms to solve well-defined challenges.

1980s: Hans Moravec's Paradox

> "It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility."

The Core Problem

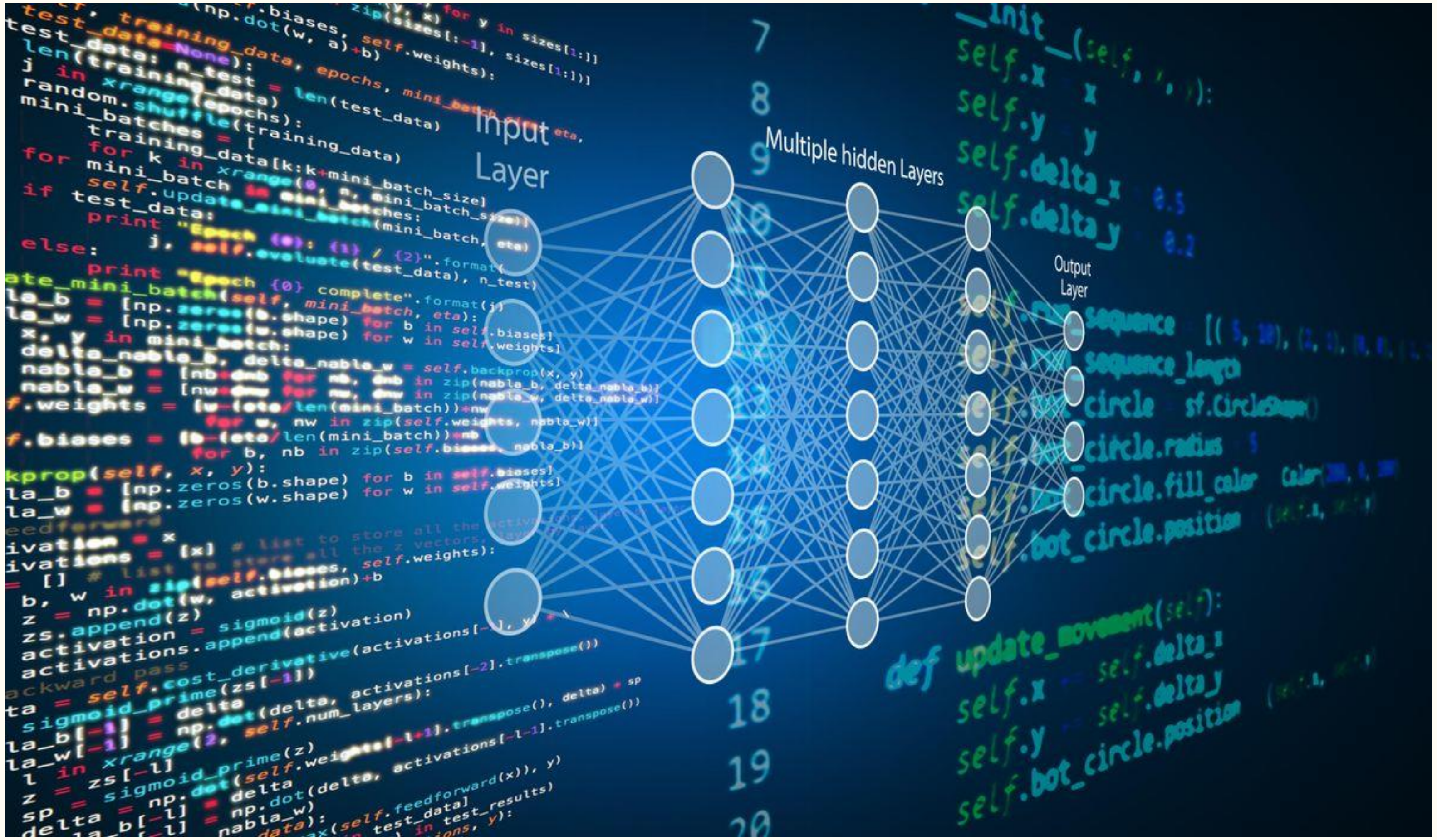
Machines couldn't learn new rules on their own.

Systems would break when encountering situations outside their programmed knowledge—they lacked the flexibility and adaptability that characterizes human intelligence.

Result: AI Winter

Disillusionment with AI's limitations led to dramatically reduced funding and research interest throughout the 1980s and early 1990s—a period known as the "AI Winter."

2010s: Statistical Machine Learning



From Symbolic AI to Statistical Machine Learning

Symbolic AI

rules are first-order drivers

(e.g., TurboTax)

Statistical Machine Learning

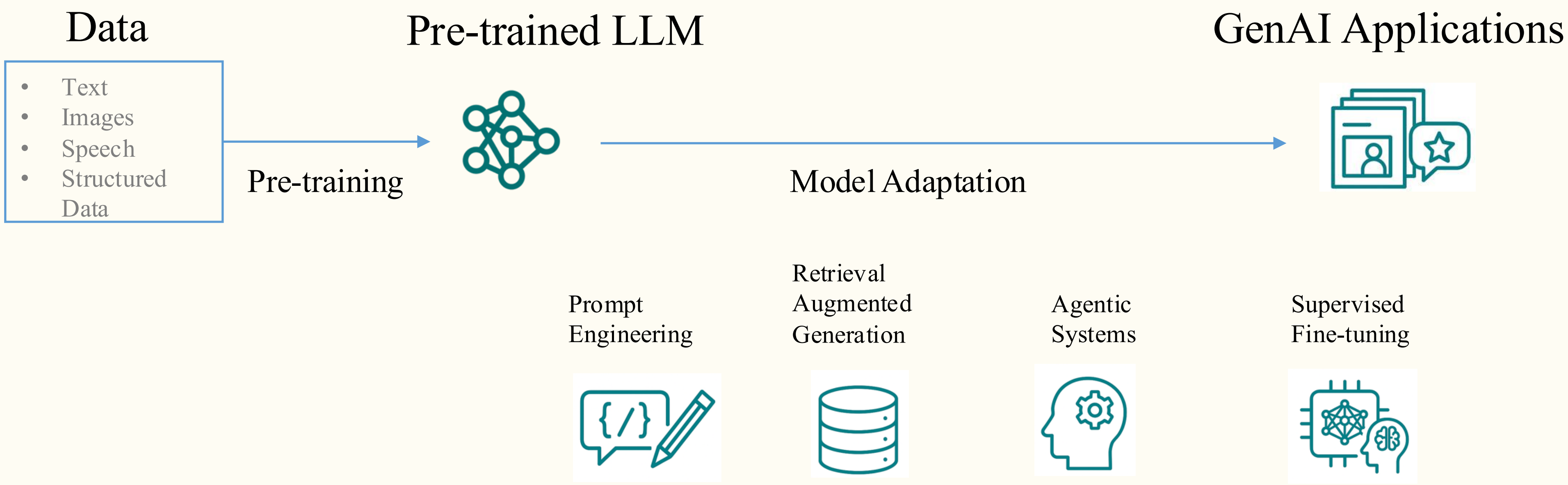
rules are second-order approximation

(e.g., ChatGPT)

What's New with GenAI?

Foundation Models

- Trained on broad data and can be adapted to many downstream tasks
- Incomplete but serves as the common basis for downstream applications



How Large Language Models Work

Text Generation: Predicting the Next Word

I love eating

bagels with cream cheese
out with friends

prompt

AI output

LLM: Autoregressive Language Model

Task: I love eating → **LLM** → ???

Unlimited possibilities

I love eating → **LLM** → bagels

I love eating bagels → **LLM** → with

I love eating bagels with → **LLM** → cream

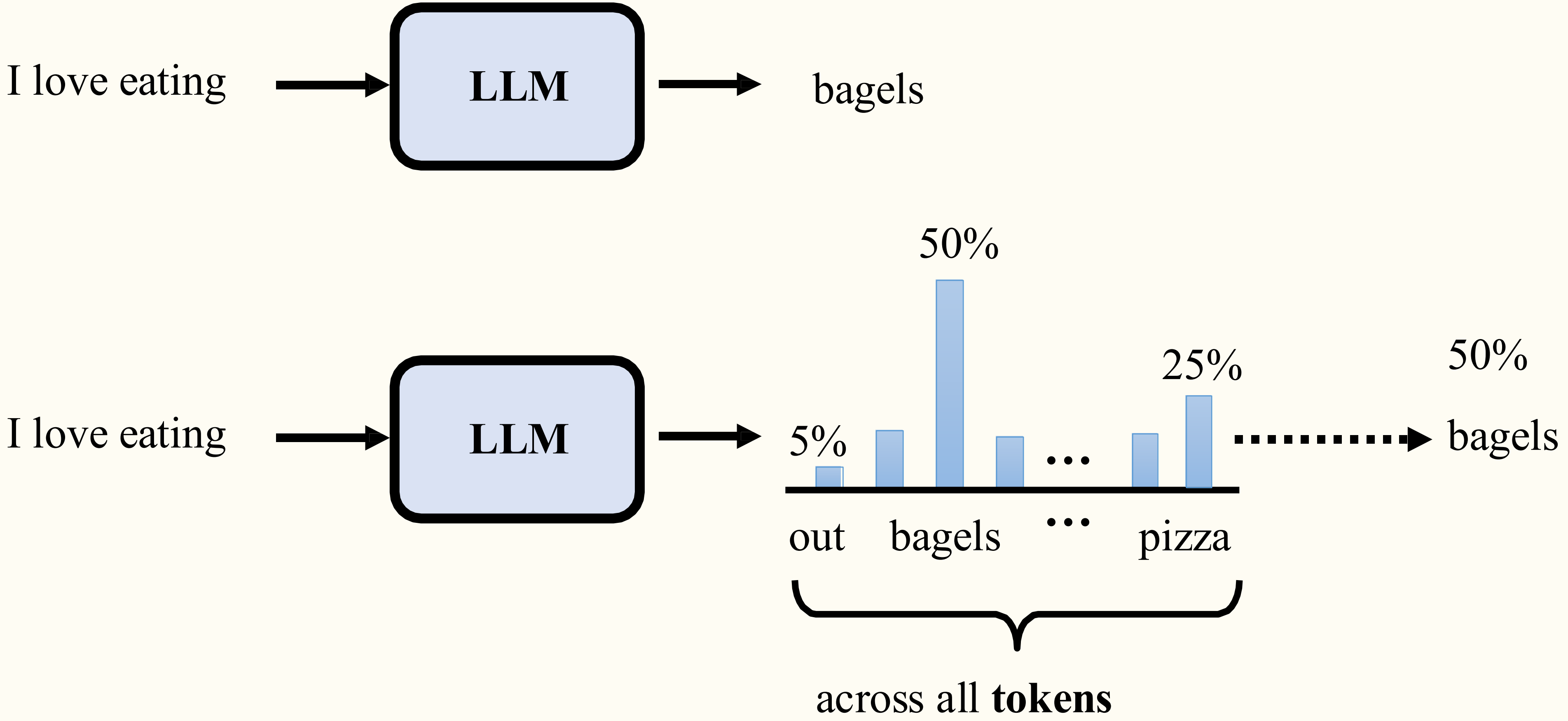
⋮

I love eating bagels with cream cheese. → **LLM** → [END]

Break it down into a series of word prediction

Limited answers

Predicting the Next Word



Transformers

Attention Is All You Need

- Tokenization
- Input Embeddings
- Position Encodings
- Query, Key, & Value
- Attention
- Self Attention
- Multi-Head Attention
- Feed Forward
- Add & Norm
- Encoders
- Masked Attention
- Encoder Decoder Attention
- Linear
- Softmax
- Decoders
- Encoder-Decoder Models

Attention Is All You Need

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Abstract

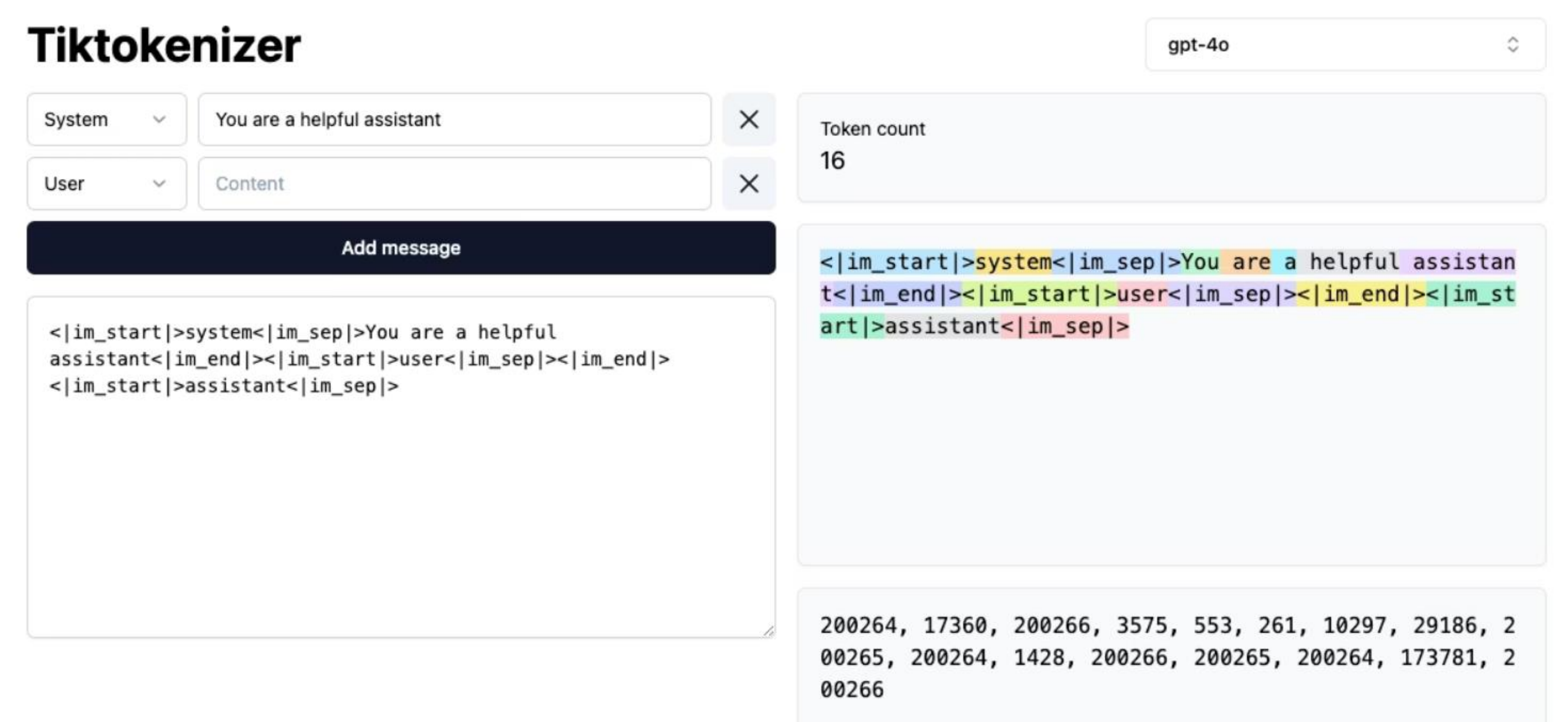
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Tokenization

Why token instead of words?

- Vocabulary Size Management
- Handling Unknown Words
- Flexibility Across Languages

The tokenization process varies between models

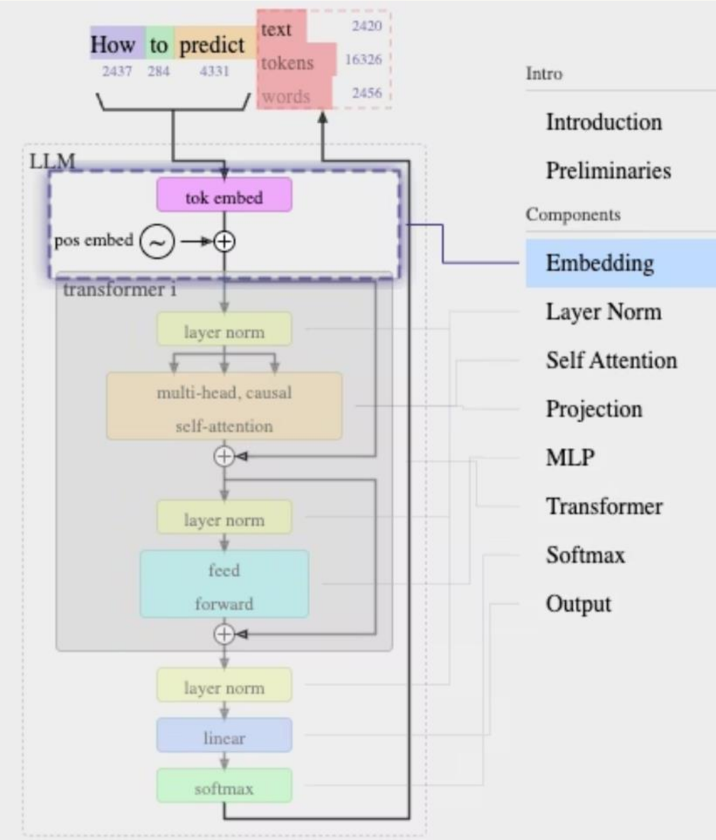


Interactive Demo: Explore how LLMs process text at https://bbycroft.net/llm

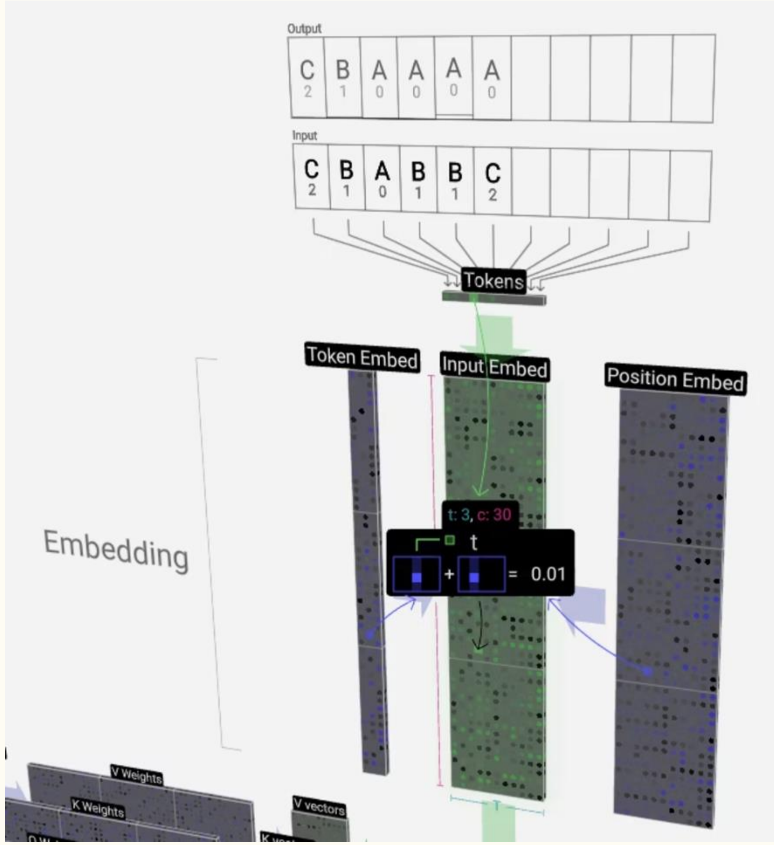
Demo: Explore tokenization interactively at tiktokenizer.vercel.app

LLM Visualization: Embedding

Tokenization



Position Encoding

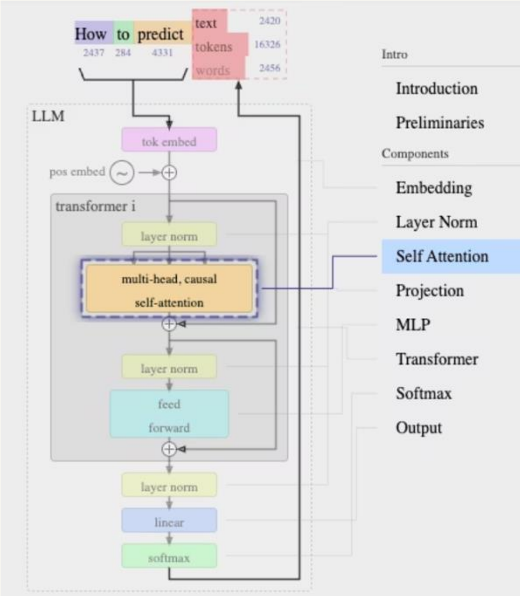


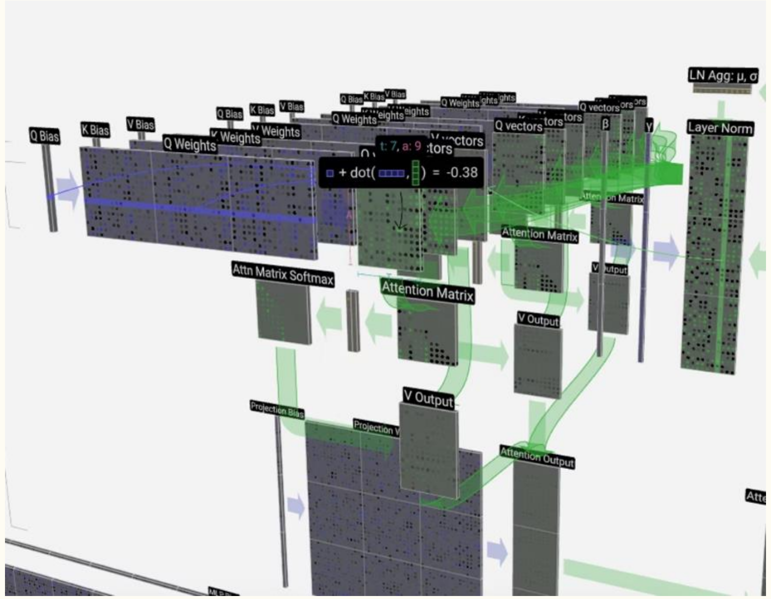
Interactive visualization: bbycroft.net/llm

Attention Mechanism

LLM Visualization: Attention

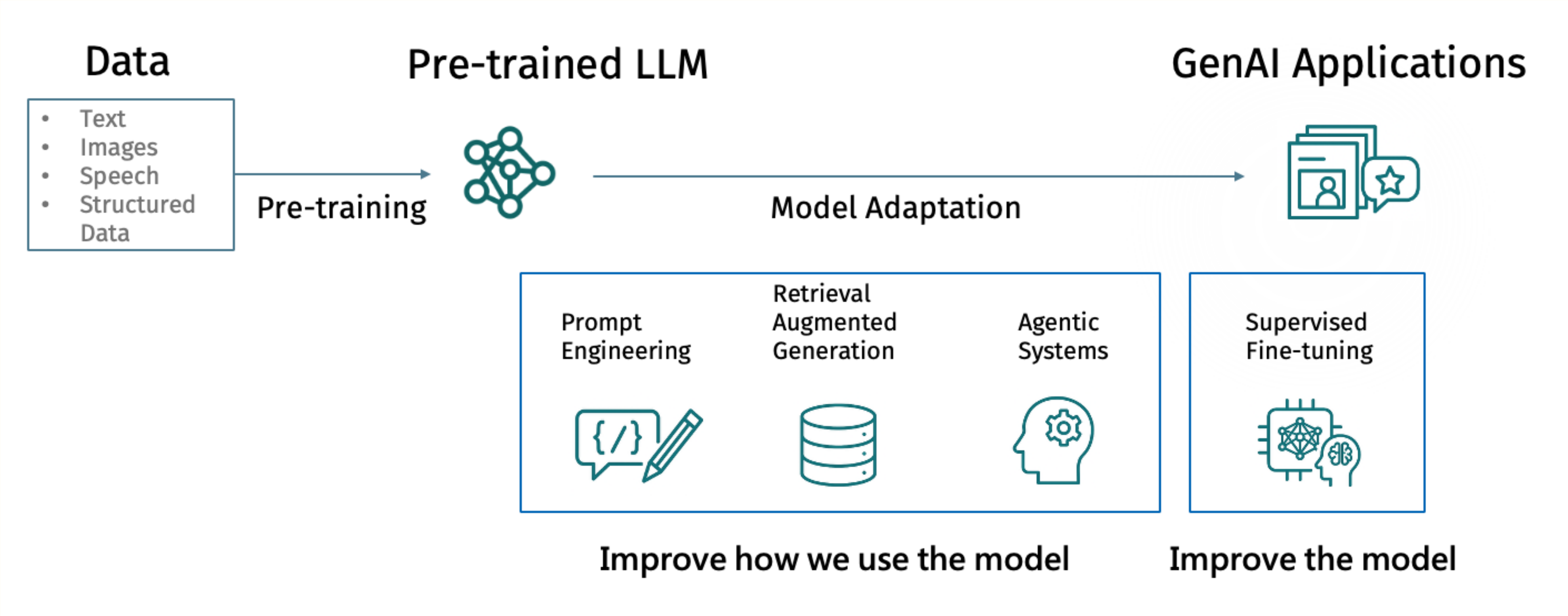
Explore: bbycroft.net/llm



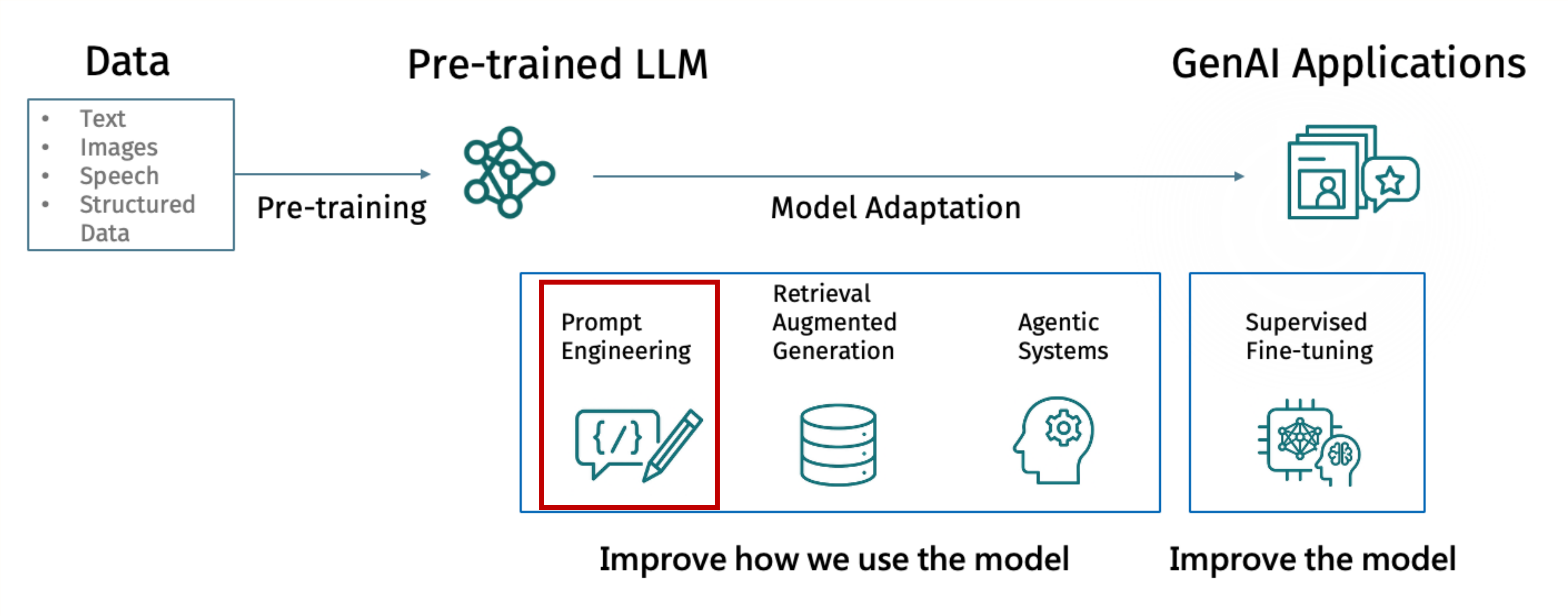


Roadmap: Customize LLMs

Customizing LLMs



Prompt Engineering



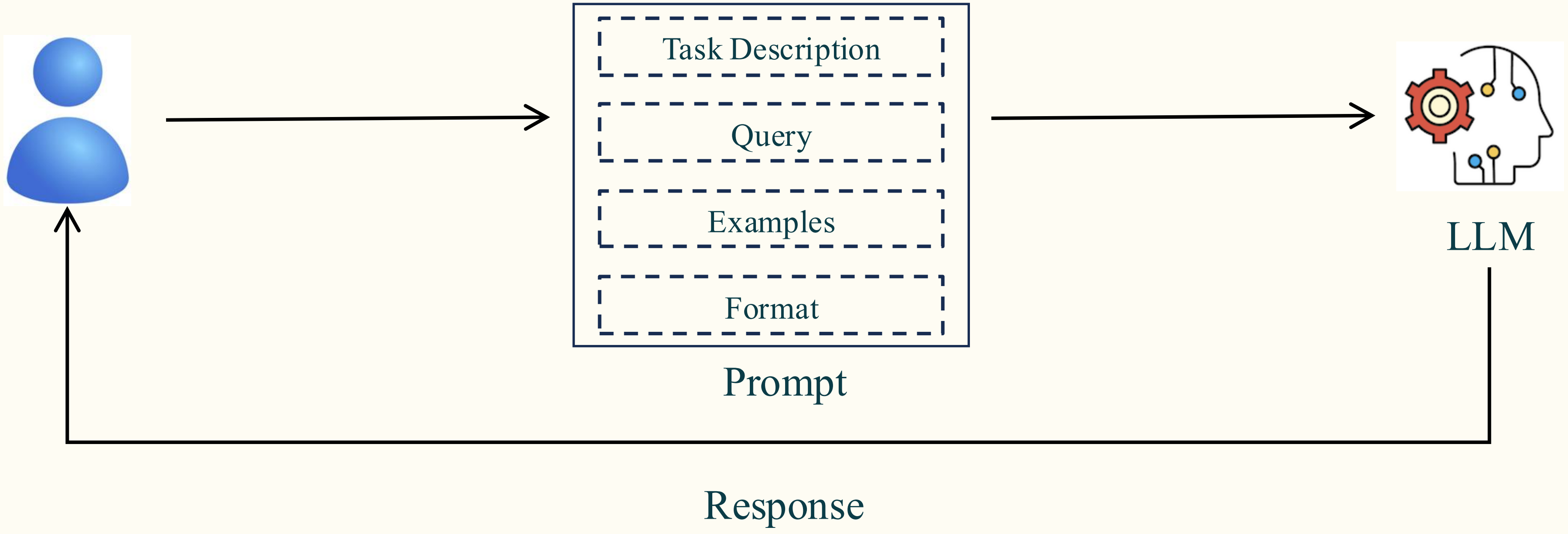
Prompt Engineering

Case: Contract Review

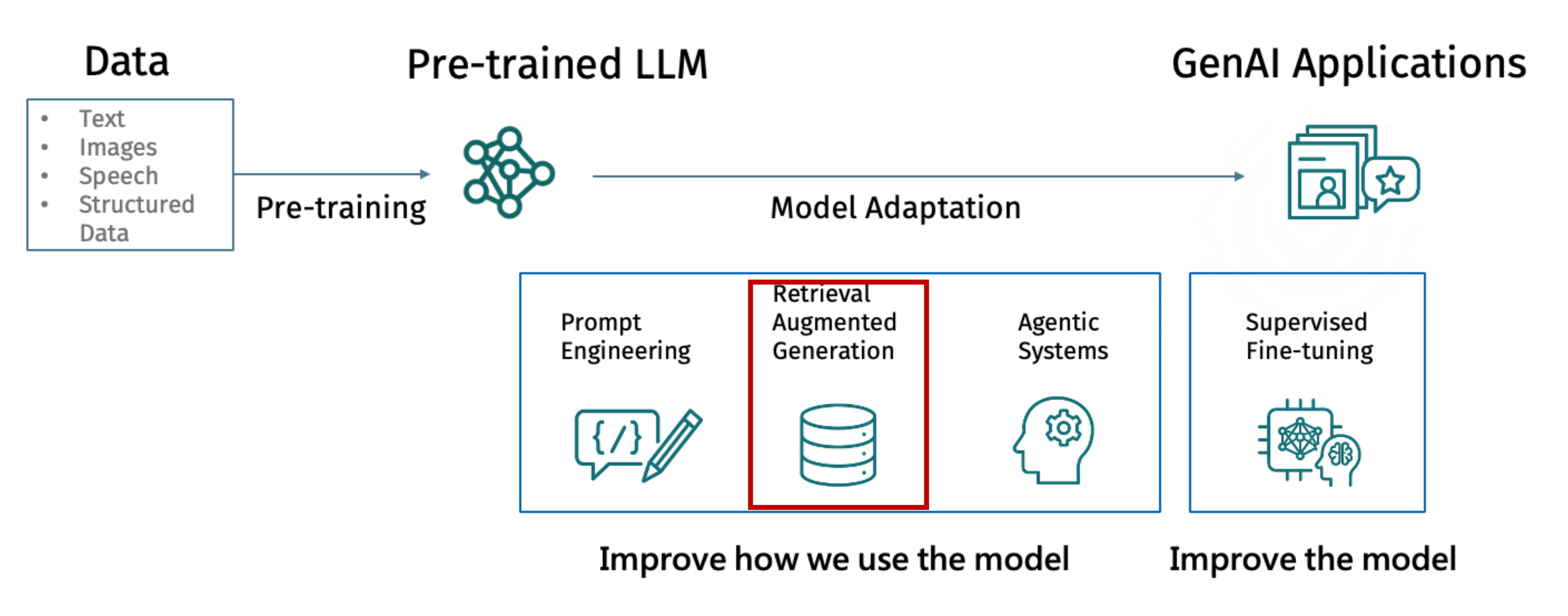
#You are an expert in contract review, especially skilled at XXXX.

#For each document， XXX

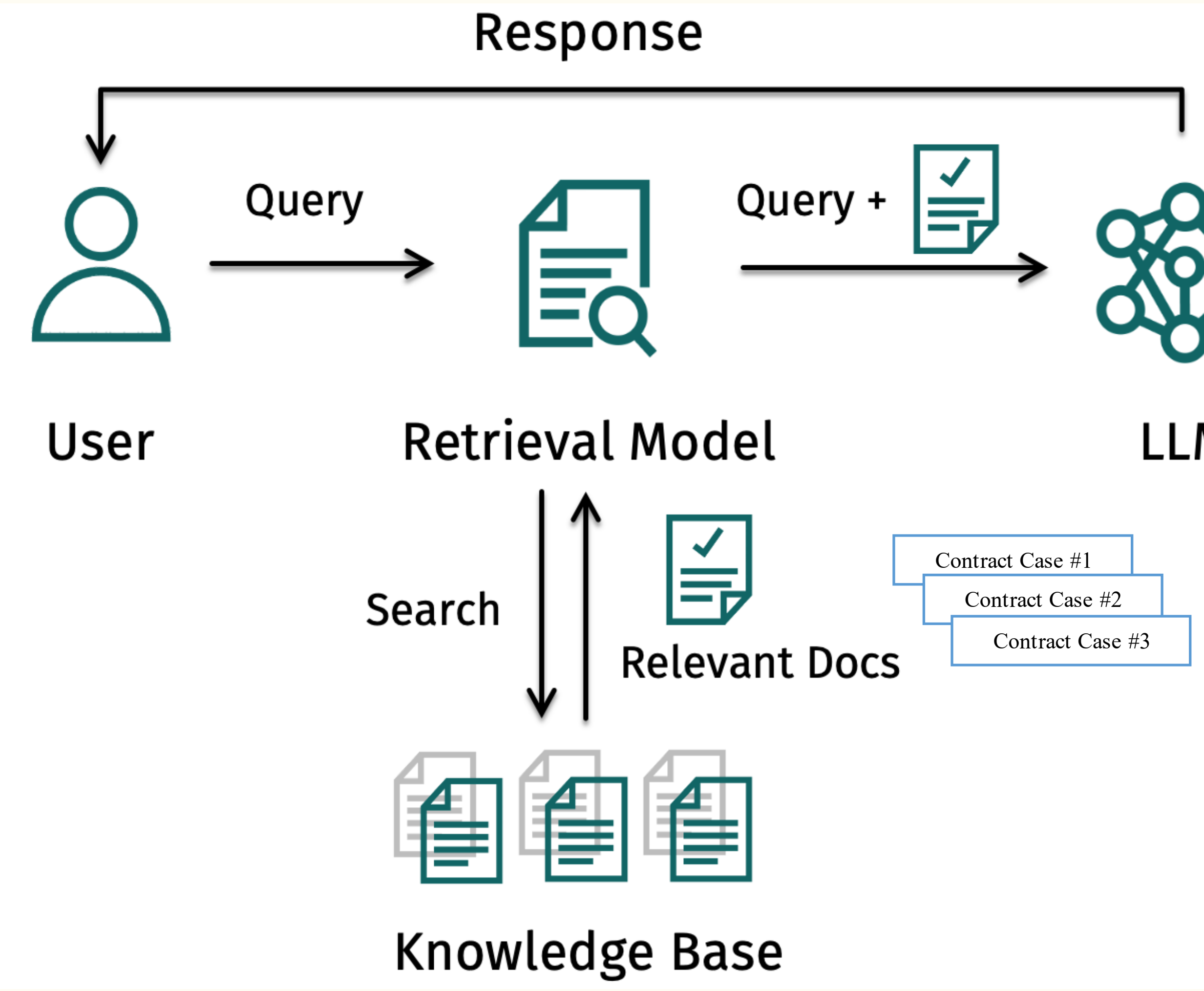
#Here are a few examples of passing and not passing contracts.



Retrieval Augmented Generation (RAG)



RAG Architecture Detail



Case: Contract Review

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